



Theory and Practice of Teaching and Learning: Rubrics, Student, and Self- Assessment

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Program Session 24

People talk (and argue) about assessment **a lot**. In fact, far too much. Almost all arguments about assessment are best ignored.

Despite this, evidence-based assessment is **critical** for learning.

I'll focus on what little we do know about what works and why.

Why assessment matters

As we saw in the scaffolding lecture, assessment is a fundamental part of the learning process because it provides feedback to learners so they can improve.

Assessment also, perhaps more critically, provides feedback to teachers to adjust instruction. In the scaffolding model, a main task of the instructor is to determine where the boundaries of the ZPD (zone of proximal development) are.

Types of assessment

Formative assessment: real time monitoring and feedback that aim to improve the mental models, skills, and problem-solving approaches of learners

Summative assessment: measurement of cumulative learning performed at the end of (or at major milestones within) instruction. Typically grades are assigned on the basis of these kinds of assessment.

Role of assessment

Formative assessment: enables learning

Summative assessment: enables credentialing

These are not the same thing. They are often at odds.

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(third type: self-assessment)

Rubrics: A Framework for HIPs

Rubrics help instructors:

Give frequent, timely, detailed, and consistent feedback

Clarify and communicate expectations

Measure progress on learning objectives and outcomes (self-assessment)

Rubrics: A Framework for HIPs

Rubrics help learners:

Understand expectations and what each component of an assignment intends to teach

Develop self-regulatory knowledge of the learning process

Qualities of good rubrics

Each part of the rubric should assess a specific and clear learning objective

Each component should explicitly state the desired traits of a solution at that score

Scoring should consistently relate to the desired traits. Scoring should account for the difference between conceptual failures and "minor mistakes".

Holistic Rubric

Points	If...
5	The student clearly understands how to solve the problem. Minor mistakes and careless errors can appear insofar as they do not indicate a conceptual misunderstanding. [a]
4	The student understands the main concepts and problem-solving techniques, but has some minor yet non-trivial gaps in their reasoning.
3	The student has partially understood the problem. The student is not completely lost, but requires tutoring in some of the basic concepts. The student may have started out correctly, but gone on a tangent or not finished the problem.
2	The student has a poor understanding of the problem. The student may have gone in a not-entirely-wrong but unproductive direction, or attempted to solve the problem using pattern matching or by rote.
1	The student did not understand the problem. They may have written some appropriate formulas or diagrams, but nothing further. Or they may have done something entirely wrong.
0	The student wrote nothing or almost nothing.

Analytic Rubric

Decomposition of motion into horizontal and vertical components	
2	Student decomposes the velocity (a vector quantity) into its vertical component v_y .
1	Student realizes that the motion should be decomposed, but does not arrive at the correct expression for v_y .
0	No attempt at decomposing the 2D motion into its vertical component.
Identification of maximum height condition	
2	Student successfully translates the physical question (the highest point of the ball) to an equation that can be used to help solve the motion ($v_y = 0$).
1	Student identifies the maximum height condition with minor mistakes.
0	Incorrect or missing identification of maximum height condition.

Rubric Activity

Discussion and share-back

Self-assessment

Formal: In addition to using assessment of students – explicit polling of students/teaching evaluations

Informal: Anonymous comment forms, office hours, informal questions

Your mission:

Using the skills that you have developed this week, you will need to lead a workshop at your home institutions in the style of the DSFP.

The workshop will include (at a minimum) a ~1 hr lecture to be followed by a “hands-on” jupyter notebook that should include problems that take ~90 min (or longer) to complete.

After:

Your workshop should incorporate learning objectives, scaffolding, and a self-assessment component.

Once you have completed the workshop, you will create a fork to the DSFP github repo `fe1lowLedWorkshops` and upload your materials (slides + jupyter notebook + solutions) in a new folder named [topic + first initial + last name]

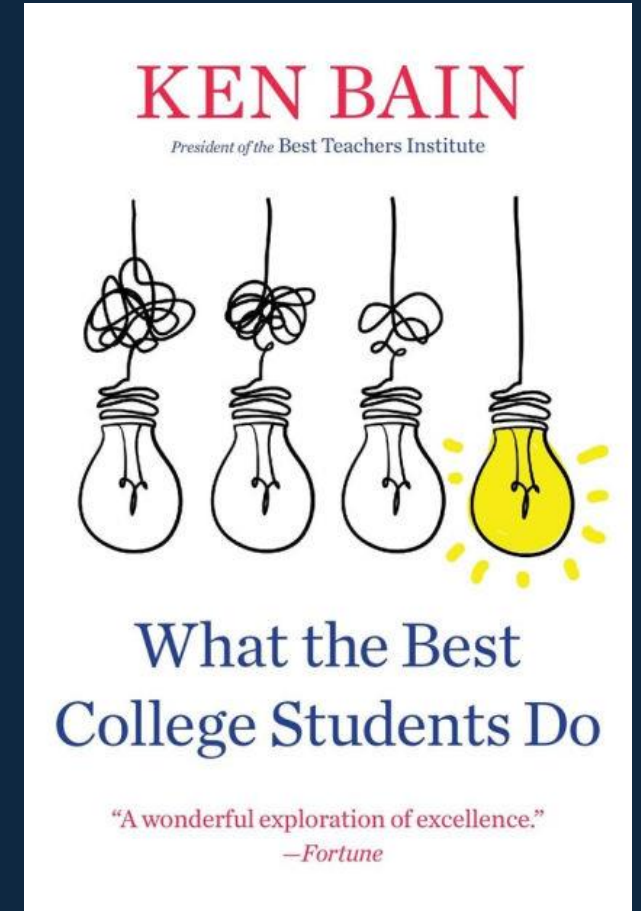
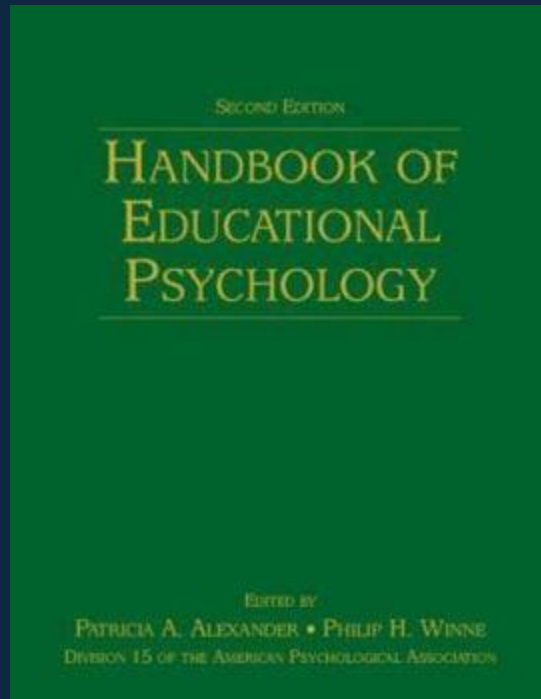
Summary

Assessment is a critical part of the learning process. Much research into improving assessment remains to be done.

Rubrics are one way of implementing best "high impact" assessment practices. Holistic and analytic rubrics offer different approaches to formulating a rubric.

Rubric elements should be clearly related to an explicit learning objective and offer a clear/measurable assessment of that objective.

References and Further Reading



References and Further Reading

